A core aspect of biological intelligence is learning to predict how the world will evolve given actions within it. In artificial systems, ‘world models’ integrate observations with actions to predict future observations. Growing in popularity for use in world models are Joint-Embedding Predictive Architectures (JEPAs), which encode an embedding of the world and predict future steps in this learned representational space. These models score well on image recognition, planning, and other benchmarks [3, 5, 9]. However, their performance has outpaced our understanding of how they select representations that support prediction. Discovering theories for how these structures self-organize during training can guide the development of more reliable and interpretable world models, and provide insight into how mammalian vision systems support visual imagination.

I will use and develop mechanistic interpretability methods to identify and characterize predictive subspaces within JEPA models. I hypothesize that a subset of JEPA representations carry information useful for causally predicting future states, and that local linearizations of the predictor should approximately map predictive subspaces at one time point into corresponding subspaces at subsequent time points. Specifically, this causal chain of representations should approximate a quiver subrepresentation – a graph-theoretic structure describing vector spaces linked by linear mappings.

Aim 1: Identify latent subspaces that causally support prediction

I will use a simulated navigation environment to study how a JEPA model represents and predicts what an agent will encounter. I will compare scenes that differ only in whether an obstacle intersects an agent’s planned trajectory, keeping the action sequence fixed. I will use SAEs to identify subspaces that shift the predicted visual sequence towards the counterfactual outcome, such as collision or passing. I can perform Centered Kernel Alignment (CKA), and employ methods similar to [8] (CCA and Hankel analysis), plus Jacobian calculations to determine directions with high local influence on predictions. I will compute Jacobian

vector-products of predictor representations and measure the resulting difference in the predicted representation to trace causal relationships. I predict that these directions should allow steering of predictor states towards possible outcome representations. I hypothesize that the subspace responsible for causal steering resembles a low dimensional manifold aligned with action space (i.e., a ring for 2D navigation), and also contains representations for more complex action motifs. These experiments will use purpose-built toy JEPA world models and larger video foundation models like V-JEPA 2 [3].

Aim 2: Characterize representation-theoretical properties of subspaces

Once I identify predictive subspaces, I can perform analysis using quiver representations [11, 15, 16]. I will create a quiver whose vertices are predictive subspaces at different times, and edges the linearization of the predictor under set action transformations. I will test whether predictive subspaces are invariant under the linearized predictor (i.e., create a subrepresentation). I will define a metric measuring invariance using residual methods [10], then perform a hypothesis test comparing predictive subspaces with random subspaces of the latent space. Testing invariance will yield information about whether the JEPA model creates predictive subspaces containing enough predictive information to be invariant under feasible actions.

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